Hints: Number theory – 1

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Bézout: let d be the least positive element of $I = \{ax + by\}$. Divide a by d with remainder; the remainder lies in I and is smaller than d, so it is 0. Hence $d \mid a$, $d \mid b$, and every common divisor divides d. Euclid's lemma: if $p \mid ab$ and $p \nmid a$, write $1 = ax + py$ and multiply by b. Uniqueness of factorisation follows by induction, cancelling a common prime.

Hint 2. CRT: the map $x \mapsto (x \bmod n_1, \ldots, x \bmod n_k)$ is a ring homomorphism. Its kernel is 0, since $n_i \mid x$ for all i forces $n \mid x$ (use Euclid's lemma). Both sides have n elements, so the map is bijective. A residue is a unit exactly when every component is a unit, so φ is multiplicative. Also $\varphi(p^e) = p^e - p^{e-1}$.

Hint 3. Euler: for $\gcd(a, n) = 1$, $x \mapsto ax$ permutes the units $r_1, \ldots, r_{\varphi(n)}$, so $\prod(ar_i) \equiv \prod r_i$; cancel. Wilson: in $\mathbb{F}_p$ only $x = \pm 1$ satisfy $x = x^{-1}$, so the other factors of $(p-1)!$ pair off into products equal to 1.

Hint 4. Converse: if $n = ab$ with $1 < a < n$, then $a \mid (n-1)!$, so $(n-1)! \equiv -1 \pmod{a}$ is impossible.

Problem 2

Hint 1. From $n \mid 2^n$, n is a power of 2, say $n = 2^m$. Then $n - 1 = 2^m - 1 \mid 2^n - 1$. When does $2^m - 1$ divide $2^N - 1$?

Hint 2. $2^m - 1 \mid 2^N - 1$ if and only if $m \mid N$. So $m \mid 2^m$, which means $m = 2^l$. Next, $n - 2 = 2(2^{m-1} - 1)$ divides $2^n - 2 = 2(2^{n-1} - 1)$, which gives $m - 1 \mid n - 1 = 2^m - 1$, that is $2^l - 1 \mid 2^{2^l} - 1$, so $l \mid 2^l$.

Hint 3. So l is a power of 2 (and $l \geqslant 1$, because $n = 2$ fails). The conditions are also sufficient. Answer: $n = 2^{2^l}$ with $l \in \{1, 2, 4, 8\}$, i.e. $n = 4, 16, 65536, 2^{256}$. The next case, $2^{2^{16}}$, is larger than 10^{100}.

Problem 3

Hint 1. By CRT, $x^2 \equiv x \pmod{n}$ splits into $x^2 \equiv x \pmod{p^e}$ for each prime power $p^e \parallel n$.

Hint 2. Modulo p^e, $x(x-1) \equiv 0$, and x, $x - 1$ are coprime, so p^e divides one of them: $x \equiv 0$ or $x \equiv 1$. Answer: $2^{\omega(n)}$, where $\omega(n)$ is the number of distinct prime factors. The idempotents of $\prod \mathbb{Z}/p^e$ are exactly the vectors with entries $0, 1$.

Problem 4

Hint 1. Write $f = ax^2 + bx + c$ and use $f(0), f(1), f(-1)$: $5 \mid c$, $5 \mid a + b$, $5 \mid a - b$, hence $5 \mid 2a, 2b$.

Hint 2. General principle: if $\deg f \leqslant p - 1$ and $p \mid f(k)$ for all k, then $\bar{f} \in \mathbb{F}_p[x]$ has p roots but degree $< p$, so $\bar{f} = 0$.

Hint 3. For $\deg f = d \geqslant p$ the statement fails: take $x^{d-p}(x^p - x)$ (Fermat). So it holds exactly when $d < p$.

Problem 5

Hint 1. $2021 = 43 \cdot 47$. Treat each prime separately. Modulo 43, by CRT, A consists of each nonzero residue mod 43 repeated 46 times.

Hint 2. So $S(j) \equiv 46 \sum_{x=1}^{42} x^j \pmod{43}$. Using a primitive root, $\sum_{x=1}^{p-1} x^j \equiv -1$ if $(p-1) \mid j$ and $\equiv 0$ otherwise. Do the same modulo 47.

Hint 3. Answer: $2021 \mid S(j)$ exactly when j is divisible by neither 42 nor 46.